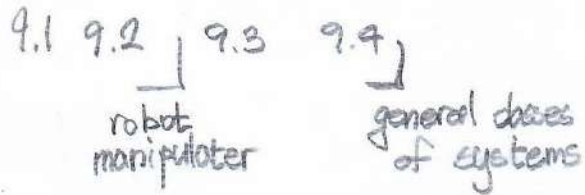


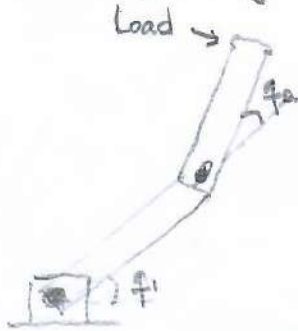
Ch 9. Control of Multi-Input Physical Systems.

- How to design a controller in a physical stance rather than mathematical.



9.1. Position & Trajectory Control.

9.2 Adaptive Trajectory Control.



$$H(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) = \tau, \quad q = [q_1, q_2]^T$$

where

$$\left\{ \begin{array}{l} H(q) \in \mathbb{R}^{2 \times 2} : \text{inertia matrix (sym \& pos. def.)} \\ C(q, \dot{q}) \dot{q} : \text{Centrifugal \& Coriolis torque} \\ g(q) : \text{gravitational torque} \\ \tau : \text{control input to each joint (control law)} \end{array} \right.$$

Q: How to construct τ to achieve $q \rightarrow q_d$.

$$\left\{ \begin{array}{l} q_d : \text{const.} : \text{position control} \\ q_d = q_d(t) : \text{trajectory control} \end{array} \right.$$

• 2 Important Notes before we start.

① $E_k = \frac{1}{2} \dot{q}^T H(q) \dot{q} \quad (\approx \frac{1}{2} m v^2)$

② $H(q)$ is uniformly pos. definite i.e. $H(q) \geq \alpha I_2$ for some $\alpha > 0$

Pf) If not, $\exists \{q_n\} \subset \mathbb{R}^2$ s.t. $\lambda_{\min}(H(q_n)) \rightarrow 0$ as $n \rightarrow \infty$.

$$\Rightarrow \exists \{q_n\} \subset \{q_n\} \text{ s.t. } q_n \rightarrow q_\infty \text{ (B-W thm)}$$

$$\Rightarrow H(q_\infty) \text{ has zero eigenvalue}$$

$$\Rightarrow \exists \dot{q} \neq 0 \text{ s.t. } E_k = \frac{1}{2} \dot{q}^T H(q_\infty) \dot{q} = 0 \quad \nabla$$

workspace is closed (compact)

4.1.1. Positional Control (q_d : const. vector)

We use a P.D controller for each joint independently.

i.e. $\tau_j = -K_{P_j} \tilde{q}_j - K_{D_j} \dot{\tilde{q}}_j$ where $\tilde{q}_j = q_j - q_{d_j}$ ($j = 1, 2, K_{P_j}, K_{D_j} > 0$)

$$\Rightarrow \tau = -K_P \tilde{q} - K_D \dot{\tilde{q}}$$

manip. on horizontal plane.

Transferring the problem to Hamiltonian form, assuming $g(q) = 0$,

$$\frac{1}{2} \frac{d}{dt} (\dot{q}^T H \dot{q}) = \frac{1}{2} \ddot{q}^T H \dot{q} + \frac{1}{2} \dot{q}^T (\dot{H} \dot{q} + H \ddot{q})$$

$$\approx \frac{1}{2} \ddot{q}^T H \dot{q} + \frac{1}{2} \dot{q}^T H \ddot{q}$$

$\dot{q}^T \dot{H} \dot{q} \approx 0$ often neglected in control context.

$$= \dot{q}^T H \ddot{q}$$

$$= \dot{q}^T (\tau - C \dot{q} - g)$$

$\dot{q}^T C \dot{q}$ neglected. (related with $\dot{H}$)

$$= \dot{q}^T \tau$$

$$= \dot{q}^T (-K_P \tilde{q} - K_D \dot{\tilde{q}})$$

pot. E ($\approx \frac{1}{2} k \tilde{q}^2$)

mech. Energy $V = \frac{1}{2} [\dot{q}^T H \dot{q} + \tilde{q}^T K_P \tilde{q}]$

$$\Rightarrow \dot{V} = \dot{q}^T (-K_P \tilde{q} - K_D \dot{\tilde{q}}) + \frac{1}{2} \frac{d}{dt} (\tilde{q}^T K_P \tilde{q})$$

$$= \dot{q}^T (-K_P \tilde{q} - K_D \dot{\tilde{q}}) + \tilde{q}^T K_P \dot{\tilde{q}}$$

$$= -\dot{q}^T K_D \dot{\tilde{q}} \leq 0$$

$$\dot{q}^T K_P \tilde{q} = \tilde{q}^T K_P \dot{\tilde{q}}$$

By Barbalat's lemma ($\dot{V}$ is bounded $\Rightarrow V: UC$), $\dot{V} \rightarrow 0$ as $t \rightarrow \infty$.

We should also check system doesn't get stuck. (goal: $q \rightarrow q_d$)

i.e. $\dot{V} = 0$ only when $\tilde{q} = 0$

Pf) $\dot{V} = 0 \Leftrightarrow \dot{\tilde{q}} = 0$

$$\Rightarrow \ddot{q} = H^{-1} \tau = H^{-1} (-K_P \tilde{q} - \cancel{K_D \dot{\tilde{q}}})$$

$$= -H^{-1} K_P \tilde{q}$$

$$\Rightarrow \dot{\tilde{q}} \text{ is identically } 0 \text{ only if } \tilde{q} = 0$$

9.1.2 Trajectory Control ($q_d = q_d(t)$)

• Feedback Linearization

$$\tau = H\ddot{q} + C\dot{q} + g \quad \text{where} \quad \ddot{q} = \ddot{q}_d - 2\lambda\dot{\tilde{q}} - \lambda^2\tilde{q} \quad (\lambda > 0)$$

$\Rightarrow H\ddot{q} + C\dot{q} + g = \tau$ turns into

$$\ddot{q} = \ddot{q}_d - 2\lambda\dot{\tilde{q}} - \lambda^2\tilde{q}$$

$\Rightarrow \ddot{\tilde{q}} + 2\lambda\dot{\tilde{q}} + \lambda^2\tilde{q} = 0$. i.e. $\tilde{q}$ exponentially converges to 0.

• Robust Control.

Aim to handle uncertainty & disturbances in the system.

We derive a robust traj. controller using sliding control. (chap 7)

sliding condition for multi-input case is

$$\frac{1}{2} \frac{d}{dt} (s^T s) \leq -\eta (s^T s)^{1/2} \quad (\eta > 0) \quad \text{where}$$

$$s = \overset{\substack{\text{velocity} \\ \text{error}}}{\dot{\tilde{q}}} + \overset{\substack{\text{sym. pos. def.}}}{\lambda \tilde{q}} \\ = \dot{q} - \dot{q}_r \quad (\dot{q}_r = \dot{q}_d - \lambda \tilde{q})$$

Note

① Assuming bdd initial cond. s : bdd $\Leftrightarrow \dot{\tilde{q}}$ & $\tilde{q}$ bdd.

② $s \rightarrow 0 \Rightarrow \dot{\tilde{q}}$ & $\tilde{q} \rightarrow 0$

③ Look at Ham. Form again with $g \neq 0$.

$$\dot{q}^T (\tau - g) = \frac{1}{2} \frac{d}{dt} [\dot{q}^T H \dot{q}]$$

$$\Rightarrow \dot{q}^T (H\ddot{q} + C\dot{q} + g - g) = \dot{q}^T H\ddot{q} + \frac{1}{2} \dot{q}^T \dot{H} \dot{q}$$

$$\Rightarrow \dot{q}^T (\dot{H} - 2C) \dot{q} = 0$$

$$\Rightarrow \dot{H} - 2C \text{ is skew-symmetric}$$

$$\Rightarrow (\dot{H} - 2C)^T = -(\dot{H} - 2C)$$

$$\Rightarrow \dot{H} = C + C^T$$

$\downarrow$ not quite obvious $\Rightarrow$ Accep.
but proof is too long (p.400).

9.2 Adaptive Robot Trajectory Control.

We deal with the situation when the manipulator's mass properties are unknown. But want $q \rightarrow q_d$.

a : unknown param. of mass properties (const. vector).

$$\tilde{a} = \hat{a} - a$$

$$V(t) = \frac{1}{2} [s^T H s + \tilde{a}^T \Gamma^{-1} \tilde{a}] \quad \text{where } \Gamma \text{ is a sym. pos. def. matrix}$$

$$\Rightarrow \dot{V}(t) = s^T (-H\ddot{q}_r - C\dot{q}_r - g) + \hat{a}^T \Gamma^{-1} \dot{\tilde{a}}$$

Since H, C, g all linearly depends on a , we can construct Y s.t.

$$H\ddot{q}_r + C\dot{q}_r + g = Y(q, \dot{q}, \ddot{q}_r) a$$

Now, take $\dot{\hat{a}} = Y^T s - \Gamma^{-1} \hat{a}$ using P.D controller,

$$\begin{aligned} \Rightarrow \dot{V} &= s^T (Y\hat{a} - K_D s - Y a) + \hat{a}^T \Gamma^{-1} \dot{\tilde{a}} \\ &= s^T Y \tilde{a} - s^T K_D s + \hat{a}^T \Gamma^{-1} \tilde{a} \end{aligned}$$

update $\hat{a}$ by $\dot{\hat{a}} = -\Gamma Y^T s$

$$\Rightarrow \dot{V} = -s^T K_D s \leq 0 : \text{ global stability \& convergence of } \tilde{q} \text{ to } 0$$

Dealing with prob.

Define $V(t) = \frac{1}{2} s^T H s$

$$\begin{aligned}\Rightarrow \dot{V}(t) &= s^T H \dot{s} + \frac{1}{2} s^T \dot{H} s \\ &= s^T H (\ddot{q} - \ddot{q}_r) + \frac{1}{2} s^T \dot{H} s \quad \downarrow \quad s = \dot{q} - \dot{q}_r \\ &= s^T (\Gamma - C\dot{q} - g - H\ddot{q}_r) + \frac{1}{2} s^T \dot{H} s \quad \downarrow \quad H\ddot{q} = \Gamma - C\dot{q} - g \\ &= s^T \{ \Gamma - C(s + \dot{q}_r) - g - H\ddot{q}_r \} + \frac{1}{2} s^T \dot{H} s \\ &= s^T (\Gamma - H\ddot{q}_r - C\dot{q}_r - g) + \frac{1}{2} s^T (\dot{H} - 2C) s\end{aligned}$$

Choose $\Gamma = \hat{\Gamma} - K \cdot \text{sgn}(s)$
 $\nearrow$ const. vector we can choose
 $\nwarrow$ element wise
 Where $\hat{\Gamma} = \hat{H}\ddot{q}_r + \hat{C}\dot{q}_r + \hat{g}$ (makes $\dot{V} = 0$ given all dynamics known)

$$\Rightarrow \dot{V} = s^T [\tilde{H}\ddot{q}_r + \tilde{C}\dot{q}_r + \tilde{g}] - \sum_i k_i |s_i| \quad (\tilde{H} = \hat{H} - H)$$

Given bounds of modelling errors $\tilde{H}, \tilde{C}, \tilde{g}$, if we choose k_i s.t.

$$k_i \geq |[\tilde{H}\ddot{q}_r + \tilde{C}\dot{q}_r + \tilde{g}]_i| + \eta_i, \text{ then } \dot{V} \leq -\sum_i \eta_i |s_i|$$

we get the desired sliding cond.

$\Rightarrow$ sliding surface $s=0$ is reached in finite time.

$\hookrightarrow q \rightarrow q_d$ exponentially.